Reviewer Pack — Minimal Rank of Noncrossed Products Bounds Communication Com…

Entry 0484d753dc65 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-29 00:03:50 UTC

Minimal Rank of Noncrossed Products Bounds Communication Complexity for Max-Cut

Entry ID: `0484d753dc65` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-29 00:03:50 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative Geometry (Noncrossed Products) **Field B** (complexity object): Communication Complexity (Max-Cut)

Statement:

> For a noncrossed product of groups G = H □ φ K, where H and K are finite groups and φ is a homomorphism from K to Aut(H), the minimal rank of the group G bounds the randomized communication complexity for Max-Cut on n vertices. Specifically, for any instance of Max-Cut with n ≤ 40 vertices, the randomized communication complexity C(n) satisfies E[C(n)] = Θ(min_rank(G)).

Rationale (proposer's reasoning):

> Noncrossed products provide a rich structure that can capture complex interactions in group theory. Their minimal rank could potentially reveal underlying structural properties that influence communication complexity, as Max-Cut is known to be related to the difficulty of distributing information among parties.

Taxonomy category: `Noncrossed_Products` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `5cc79e50b67ac9e8`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if the empirical mean of randomized communication complexities E[C(n)] for Max-Cut instances on n ≤ 40 vertices is within a constant factor of Θ(min_rank(G)) across all noncrossed products G = H □ φ K, with at least 30 seeds.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	1.00	UNCERTAIN	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    # Define groups H and K (simple examples for testing)
    H = {1, 2, 3}
    K = {4, 5}

    # Define the homomorphism φ from K to Aut(H)
    phi = {4: lambda x: x + 1, 5: lambda x: x - 1}

    # Compute the minimal rank of G = H □ φ K
    min_rank_G = len(H) * len(K)

    # Define a Max-Cut instance on n vertices (simple example for testing)
    n = 40
    max_cut_instance = [random.choice([0, 1]) for _ in range(n)]

    # Compute the randomized communication complexity C(n)
    communication_complexity = sum(max_cut_instance) * (n - sum(max_cut_instance)) / n

    return {
        "metric_name": "communication_complexity",
        "metric_value": communication_complexity,
        "instances_tested": 1,
        "conjecture_holds": abs(communication_complexity - min_rank_G) <= 0.5 * min_rank_G,
        "counterexample": "" if conjecture_holds else f"max_cut_instance={max_cut_instance}"
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
```

```
        print(f"TRIAL: {{\"seed\": {seed}, **{trial_result}}}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) /
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    else:
        first_failing_seed = next(result["seed"] for result in results if not result["conjecture_h
        print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing_
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_14e4b18a.py", line 51, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_14e4b18a.py", line 43, in run_trial
    "counterexample": "" if conjecture_holds else f"max_cut_instance={max_cut_instance}"
                            ^^^^^^^^^^^^^^^^
NameError: name 'conjecture_holds' is not defined
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test code crashed before producing data, which prevents us from verifying the conjecture's support conditions. | next: Re-run the test with proper error handling to ensure it completes and produces results.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13078
2	propose	ollama_remote	glm4:latest	0	0	8961

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
3	preregistration	ollama_remote	glm4:latest	0	0	5699
4	novelty	ollama_remote	glm4:latest	0	0	4869
5	novelty	ollama_remote	glm4:latest	0	0	5081
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	21110
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10868
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8187
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7264
10	judge	ollama_remote	glm4:latest	0	0	10270

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 95386 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/0484d753dc65.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/0484d753dc65.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/0484d753dc65.tar.gz` (if generated)